

Micro Set 1. DD/SS — Model Answer Scheme

How to use this scheme. Each answer is in A-grade student-script style with mark allocations shown. Examiner notes flag common errors.

Question 1

[2 marks]

The URA Private Property Price Index rose persistently throughout 2018–2023, from 149.6 to 200.7, an increase of approximately 34 per cent. *[1 mark for direction.]* The rate of increase was uneven: relatively gradual between 2018 and 2020 (149.6 to 157.0, +5.0%) but sharply accelerated between 2020 and 2022 (157.0 to 187.8, +19.6%) before slowing in 2023. *[1 mark for identifying the inflection.]*

Examiner Note. 2-mark “describe” questions reward identifying both the overall direction and the inflection. Stating “it rose” alone caps at 1 mark.

Question 2

[2 marks]

Equilibrium price is the price at which the quantity demanded equals the quantity supplied in a market, with no tendency for further change. *[1 mark.]*

Extract 1.1 identifies several demand-increasing factors in 2021–2022 (returning expatriates, foreign-buyer interest, low mortgage rates) coinciding with a slow supply response (completions fell to 6388 units in 2021). At the original equilibrium price, demand exceeded supply, generating upward pressure on price until a new, higher equilibrium was reached. *[1 mark for application.]*

Examiner Note. The definition must include the equality of Q_d and Q_s and the no-further-change condition. A vague “where supply meets demand” loses precision.

Question 3

[4 marks]

Resident population growth and property prices.

Resident population growth swung from -0.7% in 2021 to $+4.2\%$ in 2023, the highest rate in over a decade (Extract 1.1, Table 1.1). *[1 mark for data anchor.]*

A larger resident population requires more housing units, raising the quantity of private residential property demanded at every price level. In demand-supply terms, the demand curve shifts rightward. *[1 mark for DD-shift mechanism.]*

Two channels reinforce this. First, direct housing demand — returning Singaporeans and re-entering expatriates require accommodation immediately. Second, indirect demand — higher population growth raises rental demand sharply (the Median Rental Price Index rose from 99.9 in 2020 to 153.6 in 2023, a 54% increase, Table 1.1), and elevated rents incentivise investor purchases of property as buy-to-rent assets, adding a further demand impulse. *[1 mark for dual-channel.]*

At the original equilibrium price, $Q_d > Q_s$, creating upward pressure on price until a new equilibrium at higher P and higher Q is reached. *[1 mark for equilibrium-adjustment language.]*

Examiner Note. Strong 4-mark answers explicitly use the price-adjustment mechanism (“ $Q_d > Q_s$ ” / “upward pressure on price”). Stating only “demand rose so price rose” is descriptive, not analytical.

Question 4

[4 marks]

Mortgage rates and demand for property.

A fall in mortgage rates reduces the periodic cost of servicing a property loan. For a buyer financing a 30-year mortgage at the typical loan-to-value ratio of 75%, a fall from 3.1% to 1.3% reduces monthly debt-service costs by roughly 25%. *[1 mark for affordability mechanism + magnitude.]*

Lower debt-service costs raise the maximum property price that any given household income can afford under the Total Debt Servicing Ratio (TDSR) framework. Households previously priced out at the higher rate become eligible buyers, and existing buyers can bid for higher-priced properties. The demand curve shifts rightward. *[2 marks for TDSR mechanism + DD shift.]*

Additionally, the opportunity cost of holding fixed-income savings (deposits, bonds) at lower rates falls relative to the expected return on property as an investment asset. This raises investor demand alongside owner-occupier demand, amplifying the rightward demand shift. *[1 mark for portfolio-substitution channel.]*

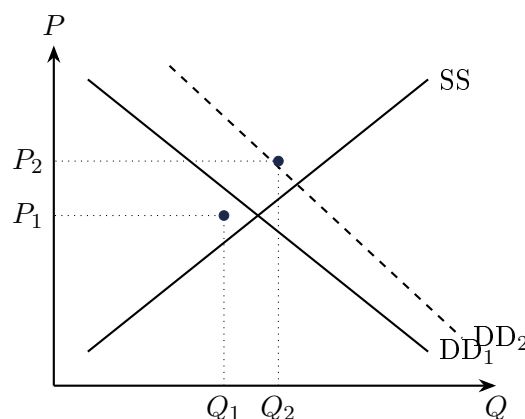
Examiner Note. Top-mark answers identify *both* the financing channel (lower repayments) *and* the portfolio channel (lower opportunity cost). Single-channel answers earn 2–3 marks.

Question 5

[6 marks]

Population + slow supply → rapid price appreciation.

Extract 1.1 identifies a simultaneous demand shock (population growth, foreign-buyer interest, low rates) and supply constraint (completions fell from 9112 to 6388 between 2018 and 2021, an exceptional contraction).



In the diagram, the rightward demand shift from DD_1 to DD_2 reflects the combined demand impulse. With supply (SS) relatively fixed in the short run, the equilibrium price rises sharply from P_1 to P_2 and quantity rises modestly from Q_1 to Q_2 . *[3 marks for diagram fully labelled with axes, both curves, shifted equilibrium.]*

The slow supply response reflects the fundamentally inelastic short-run supply of housing: development cycles take 3–5 years, construction labour and materials are constrained, and the URA's release of state land follows a planning timeline. Therefore, even after demand surged, the supply curve did not shift rightward meaningfully until 2023, when deferred projects completed simultaneously (19968 units, Table 1.1). *[2 marks for PES analysis + supply pipeline.]*

The combination of inelastic supply and large demand shift explains the unusually large price effect: when PES is low, a given demand shift translates into a large ΔP and small ΔQ , the textbook pattern observed in 2021–2022. *[1 mark for elasticity-based magnitude conclusion.]*

Examiner Note. The strongest answers explicitly invoke the price-elasticity-of-supply (PES) as the reason why the price effect was so large. Without this, the explanation is descriptive rather than analytical.

Question 6

[6 marks]

ABSD increase and the demand–equilibrium-price effect.

The Additional Buyer's Stamp Duty (ABSD) on foreign buyers rose from 20% (2018) to 60% (2023), a tripling of the effective transaction cost of foreign-buyer purchases. *[1 mark for data anchor.]*

ABSD operates as a tax on property purchase by foreign buyers, raising the effective price they must pay above the seller's listed price. For any given property price P , foreign buyers' demand falls because the all-in cost ($P + \text{ABSD}$) rises. This shifts the foreign-buyer component of demand leftward. *[2 marks for tax-on-demand mechanism.]*

Extract 1.1 confirms the empirical impact: foreign-buyer transactions fell by approximately 60% in 2023, and their share of total transactions fell from 6.0% to 1.6%. *[1 mark for evidence anchor.]*

The aggregate demand curve in the property market is the horizontal sum of resident and foreign-buyer demand. With foreign-buyer demand reduced sharply, total demand shifts leftward. The new equilibrium price is lower than it would have been absent the ABSD increase, even though the total price level continued to rise in absolute terms (200.7 in 2023, Table 1.1). *[1 mark for aggregate-demand mechanism.]*

The ABSD therefore did not reverse the price trend (population growth remained dominant) but moderated its pace, with the price index continuing to rise but more slowly. *[1 mark for relative-effect framing.]*

Examiner Note. The distinction between absolute level and rate of growth is critical. ABSD did not cause prices to fall; it caused prices to rise more slowly than they would have. Failing to make this distinction caps at 4 marks.

End of Set 1 — Model Answers